INTERNSHIP REPORT

Submitted

For the Partial Fulfilment of the Requirement for the Degree of Bachelor of _____ with _____ (Major and Minor Subjects) under **Magadh University**

Submitted to

The Department of _____, _____ **(College Name)**

[Paste College Logo here]

Submitted by

Name : _____

Semester : _____

Programme FYUGP : _____

University's Registration No. : _____

College Roll No. : _____

Year : _____

RK NEXORA
CAMPUS TO CAREER
A Unit of RKNexora Private Limited.

INTERNSHIP REPORT

Topic - Educational Technology (EdTech)

Submitted to the Head of Department, _____ (College Name), in partial fulfilment of the requirements for the Degree of _____, Magadh University

Name of Student : _____

Programme : _____

University Registration No. : _____

College Roll No. : _____

Internship Registration No. : _____

Name of the IPO / Agency : RKNEXORA PRIVATE LIMITED

Internship Period : _____ to _____

Semester : _____

Year / Session : _____

RKNEXORA PRIVATE LIMITED
RKNEXORA
PRIVATE LIMITED
GAYA, BIHAR - 805236, INDIA

_____ _____

Signature of Student **Seal & Signature of IPO**

ACKNOWLEDGEMENT

I express my sincere gratitude to RK NEXORA, RKNEXORA PRIVATE LIMITED, _____ (College Name), my faculty mentors and the internship supervisor, _____, for their guidance and support. The structured activities, learning resources and regular feedback helped me complete this internship in Educational Technology (EdTech) successfully.

I am thankful to the Head of the Department and the college mentor for their encouragement and for guiding me in preparing this report. I am also grateful to the management of RKNEXORA PRIVATE LIMITED for providing a professional learning environment and the opportunity to learn and apply my knowledge in a practical setting. The concept sessions, hands-on tasks with sample data and case studies, and the supervisor's patient and detailed feedback helped me understand how technology is used to design, deliver and improve teaching and learning.

Finally, I thank my family and peers for their constant encouragement throughout the programme.

_____

Name: _____

University Registration No. - _____

INTERNSHIP COMPLETION CERTIFICATE

The Internship Completion Certificate issued by RKNEXORA PRIVATE LIMITED for the internship in Educational Technology (EdTech) is to be attached on this page.

[Attach the original / printed copy of the QR-verified Internship Completion Certificate here]

Certificate No. _____

CONTENTS

Acknowledgement..... 3

Internship Completion Certificate..... 4

Preface..... 6

Executive Summary..... 7

About the Agency..... 8

Plan and Execution..... 9

Learning Outcome..... 23

Conclusion..... 25

Attendance Sheet..... 27

PREFACE

Internships are an important part of the undergraduate programme, as they allow students to see how classroom learning is applied in real organisations. The Four-Year Undergraduate Programme (FYUGP), introduced in line with the National Education Policy 2020, gives special importance to experiential learning, skill development and employability. This report has been prepared as part of the internship requirement of Magadh University and the Higher Education guidelines applicable in Bihar.

The report describes the internship undertaken in Educational Technology (EdTech) at RKNEXORA PRIVATE LIMITED, the plan followed, the work carried out, and the learning gained. Education is changing. A student in a village of Bihar can now watch a lecture recorded in another city, take a quiz on a phone, submit an assignment online and receive feedback the same day. A teacher can see which topic the class finds difficult before the next lesson begins. This change is the work of educational technology, or EdTech: the use of digital tools, content and data to make teaching and learning more effective, accessible and engaging. It is also a fast-growing field with a demand for people who understand both learning and technology. This report describes how the main areas of EdTech were studied, and how a small online course, with an LMS structure, a working quiz page, video and content plans and a learning analytics study, was designed using dummy data.

The report has been arranged in the order suggested by the University, beginning with the cover page and acknowledgement and ending with the conclusion and attendance sheet. It is hoped that it gives a clear and honest account of the internship experience. All examples in this report use dummy data and fictitious names; no real personal or business information has been handled.

EXECUTIVE SUMMARY

This report documents the internship completed in Educational Technology (EdTech) at RKNEXORA PRIVATE LIMITED (RK NEXORA - 'Campus to Career'). The programme required a minimum of 120 hours, completed over about four to five weeks (28 working days) with daily sessions of four to five hours, and was supported by an attendance log, supervisor guidance and review.

Background. Educational technology is the study and ethical practice of facilitating learning and improving performance by creating, using and managing suitable technological processes and resources. Its areas include learning management systems (LMS), online class tools, e-content and video lectures, online assessment, gamification, learner engagement, learning analytics and course design. Good EdTech starts from the learning need and not from the tool: the technology serves the teaching, and not the other way round.

Objectives. The internship aimed to (i) understand the EdTech landscape and its trends, (ii) learn how a learning management system is set up and used, (iii) use online class tools, (iv) create e-content, (v) build online assessments, (vi) use gamification, (vii) apply engagement methods, (viii) plan and produce video lectures, (ix) read learning analytics, and (x) design a course from outcomes to assessment.

Work carried out. Each topic had a concept session followed by practice. The student designed a dummy course, Digital Literacy Basics, of eight modules, built its structure for an LMS, wrote lesson content, built a working quiz web page, planned and scripted video lessons, designed gamification elements and analysed a dummy set of learner data of 80 learners to find where learners dropped out and which quiz questions were too hard. The last part of the programme was devoted to the project, a case study exercise, a mock practical exercise, supervisor review and preparation of this report.

Project. The work was tied together in a project titled **Digital Learning Content & LMS Project**. The project designed an eight-module course for 80 dummy learners, built a working quiz page, analysed completion and quiz data (52.5% completion, mean quiz score 6.8 out of 10) and compared a gamified and a non-gamified group (60% against 45% completion).

Method. A blended method was followed - learn, practise, document and review - supported by a daily attendance log with day-wise remarks and regular feedback from the supervisor.

Learning gained. The student gained a clear understanding of the EdTech landscape and of how an LMS works, practical skill in e-content creation, assessments, video lesson planning, gamification and engagement, the ability to read learning analytics, and the habit of designing from the learning outcome.

Key observations. Technology does not teach by itself - clear outcomes, good content and timely feedback do; short, interactive lessons keep learners better than long lectures; data should be used to help learners and not to label them; and access and inclusion must be planned from the start, since many learners use only a phone and low bandwidth.

Conclusion. The internship met its objectives and contributed to the student's academic understanding, employability and confidence. The report follows the University's suggestive format: cover page, acknowledgement, certificate, contents, preface, executive summary, about the agency, plan and execution, learning outcome, conclusion and attendance sheet.

ABOUT THE AGENCY

RK NEXORA, a unit of RKNEXORA PRIVATE LIMITED, provides structured internship and skill-development programmes under the theme 'Campus to Career'. The organization focuses on academic-industry alignment, practical exposure, assessment, documentation and employability-oriented learning.

Particulars	Details
Name of Agency	RKNEXORA PRIVATE LIMITED
Programme / Unit	RK NEXORA - Campus to Career
Registered Office	C/O Mukesh Kumar, Khiri, PS Atri, Jethiyan, Atri, Gaya - 805236, Bihar, India
CIN	U85499BR2026PTC086757
Email	rknexorapvtltd@gmail.com
Website	www.rknexora.org
Internship Topic	Educational Technology (EdTech)

1. Vision

To support students in developing practical, professional and industry-relevant capabilities.

2. Mission

To deliver structured learning experiences that connect academic knowledge with workplace expectations.

3. Activities of the Agency

1. Structured internship programmes aligned with university and higher-education requirements.
2. Skill-development learning modules, practical activities and assignments.
3. Attendance tracking, supervision, assessment and documentation.
4. Employability-oriented guidance connecting campus learning to career readiness.

4. Programme Structure

The internship programme is organised around a fixed number of learning hours, completed over a defined period. Each intern is registered on the internship portal with a unique registration number, is assigned a supervisor, follows a day-wise plan and maintains an attendance log. Learning is organised into three phases: **Orientation** (Days 1-2), **Learning and Practice** (Days 3-22, ten sub-topics of two days each: one concept day and one hands-on day) and **Project and Review** (Days 23-28), followed by a concluding review.

5. Role of the Supervisor

The internship supervisor assigned by the agency guided the student through the programme, explained concepts, assigned practice tasks, reviewed the daily log and gave feedback. The supervisor also verified attendance and reviewed the student's work at the end of the programme.

PLAN AND EXECUTION

1. Objectives

The internship was planned around the following objectives, each linked to a part of the programme:

1. Understand the EdTech landscape and learning theories in brief (Week 1).
2. Learn the working of a learning management system (Week 1-2).
3. Use online class tools and create e-content (Week 2).
4. Build assessments, gamification and engagement activities (Week 2-3).
5. Plan and produce video lectures (Week 3).
6. Read learning analytics and design a course (Week 3-4).
7. Complete the project, case study and report (Weeks 4-5).

2. Methodology

The programme followed a blended method made up of four connected steps, repeated for each topic:

Learn. Each topic began with a concept session in which the supervisor explained the idea, its main terms and how it is used in real organisations. Notes were prepared in the student's own words.

Practise. The concepts were then applied in guided hands-on tasks - designing a course and its modules, writing lesson content and quiz questions, building a quiz page, writing a video script and storyboard, designing badges, calculating completion and difficulty indices and drawing charts. Only sample data and fictitious examples were used, so that no real personal or business information was handled.

Document. Every day's work was recorded in the internship attendance log with a short remark, and sheets, files and screenshots were saved so that the work could be reviewed.

Review. The supervisor reviewed the work, pointed out errors and suggested improvements. Review days, a case study and a mock practical exercise helped to consolidate the learning.

3. Understanding Educational Technology (EdTech) - Concepts Studied

3.1 The EdTech landscape

EdTech covers a wide range of tools and services. They may be grouped as learning management systems (Moodle, Google Classroom), live-class and video conferencing tools, content and video platforms, open educational resources (such as the government's DIKSHA and SWAYAM platforms), mobile learning apps, adaptive learning systems that adjust the material to the learner, tools for assessment and proctoring, simulations and virtual and augmented reality, and, increasingly, artificial intelligence for tutoring and feedback. Current trends include mobile-first learning, short videos and micro-learning, blended learning that mixes online and classroom work, personalised learning, gamification, learning analytics and the use of AI tools. The National Education Policy 2020 also stresses technology for access, teacher training and inclusion. Important issues are the digital divide (many learners lack a device or good internet), the quality of content, data privacy, screen time and the need for teacher guidance. A learner's progress depends on a few well-known ideas from learning theory: that learners build understanding actively (constructivism), that practice with feedback is essential, that people learn better in small steps and that motivation matters.

3.2 Learning management systems

A Learning Management System (LMS) is software to host courses, enrol learners, deliver content, set and grade assessments, communicate and track progress. Examples are Moodle (open source), Google Classroom, Canvas and Blackboard. The main features are the course page with sections or modules, enrolment and user roles (administrator, teacher, learner), content types (files, links, video, pages), activities (quizzes, assignments, forums), a gradebook, completion tracking, messages and announcements, and reports. A course is set up by creating the course, organising it into modules, adding the lessons and activities, setting completion rules (for example, 'view the video and score 60% on the quiz'), enrolling learners, testing the course as a learner and then opening it. The structure of the dummy course is shown below.

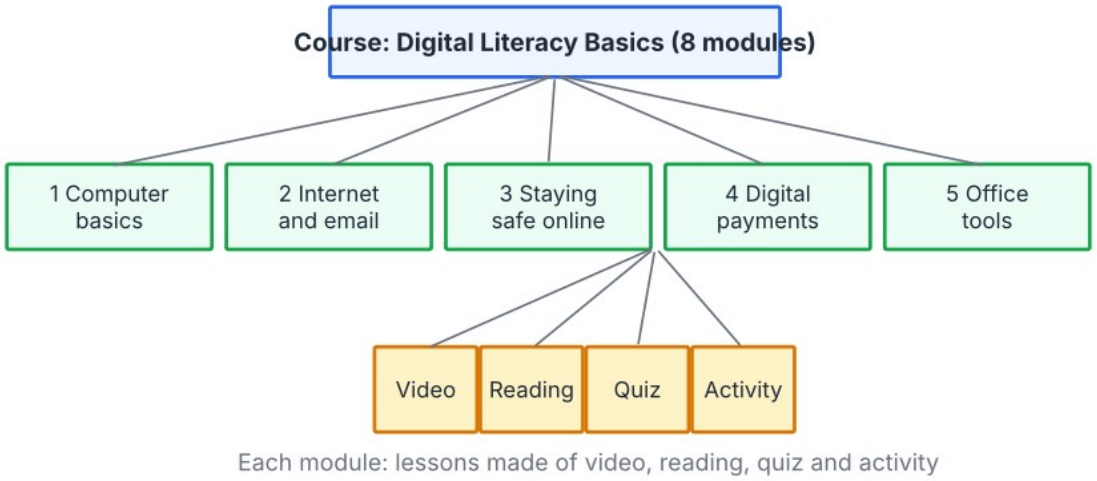


Figure 1: Structure of the dummy course in an LMS

Setup step	What was done in the dummy course
Course creation	Title, short description, duration (4 weeks) and language (Hindi and English)
Structure	Eight modules with 3 to 4 lessons each; a short quiz after every module
Completion rule	A module is complete when all lessons are viewed and the quiz is passed at 60%
Enrolment	80 dummy learners added by a list; self-enrolment closed after the first week
Communication	A welcome message, a weekly announcement and a discussion forum for doubts
Testing	The course was checked as a learner, on a phone and on a laptop, before opening

3.3 Online class tools

Live online classes use video conferencing tools such as Google Meet, Zoom and Microsoft Teams, with features such as screen sharing, a digital whiteboard, chat, polls, breakout rooms, hand-raising and recording. Good practice for a live class is to test the audio and the link beforehand, to start on time with a short plan on the screen, to keep the sessions short (45 minutes or less, with a break), to involve learners every ten minutes with a question or a poll, to record the class for those who missed it, to take attendance, and to share the materials afterwards. A few learners may be on slow connections, so a low-bandwidth option, such as audio only and slides shared in advance, should be offered. Learners should be asked to keep the microphone muted when not speaking, and the teacher should guard the privacy of the class by controlling the link and the recording.

3.4 E-content creation

E-content is digital learning material: slides, notes, infographics, videos, audio, interactive exercises, simulations and quizzes. Good e-content is organised around a clear outcome, is divided into small units of five to ten minutes (micro-learning), is written in simple language, uses visuals that support the text, gives chances to practise, has accessible design (readable font size and contrast, captions for videos, descriptions for images) and respects copyright by using original material, open licences such as Creative Commons, or permission. For slides, one idea per slide, at most a few lines, large fonts and clear images are the rules. The student prepared the lesson on 'Staying safe online' in the structure shown below.

> **Lesson plan for a micro-lesson (dummy course)**
> Lesson 3.2: Spotting a phishing message (7 minutes)
> Outcome: After this lesson the learner can identify three warning signs of a phishing message and say what to do.
> 1. Hook (1 min): a real-looking SMS 'Your account is blocked' - what would you do?
> 2. Explain (2 min): urgency, a link or number to call, a request for PIN or OTP.
> 3. Example (2 min): two messages side by side - one real, one fake.
> 4. Practise (1 min): tick the warning signs in a third message.
> 5. Check (1 min): 3-question quiz with feedback.
> Summary: Banks never ask for PIN or OTP. Do not click; call the official number.

3.5 Assessment tools

Assessment in online learning has two purposes: formative assessment, which helps learners learn through quick feedback during the course, and summative assessment, which measures achievement at the end. Tools include online quizzes with multiple choice, true or false, matching and short answer questions, assignments with rubrics, peer review and discussions. A good multiple-choice question has a clear stem, one correct answer and plausible wrong options based on common mistakes, avoids 'all of the above', and gives feedback that explains the answer. Rubrics state the criteria and levels of performance, so that learners know what is expected. Online assessments need safeguards - question banks with random selection, time limits, and, where necessary, supervised tests - but the best protection is to test understanding and not memory. The student built a working quiz page for Module 3, a practice quiz with instant feedback and badges, as shown in the following screens.

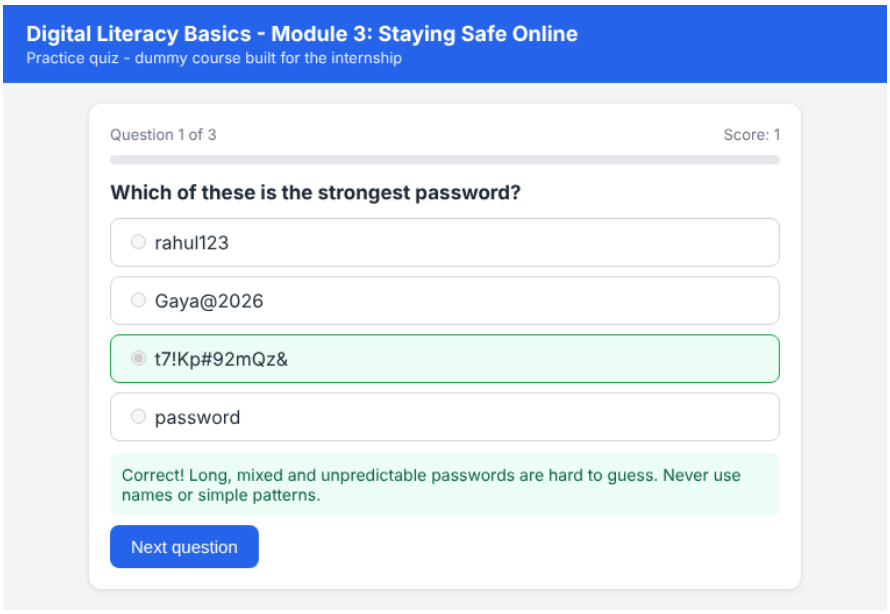


Figure 2: A correct answer receives instant feedback

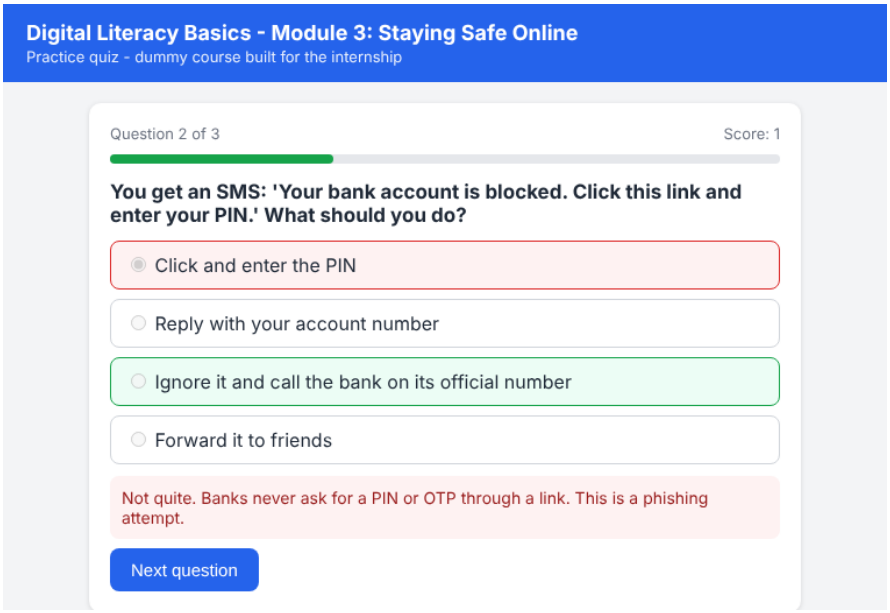


Figure 3: A wrong answer shows the correct option and an explanation

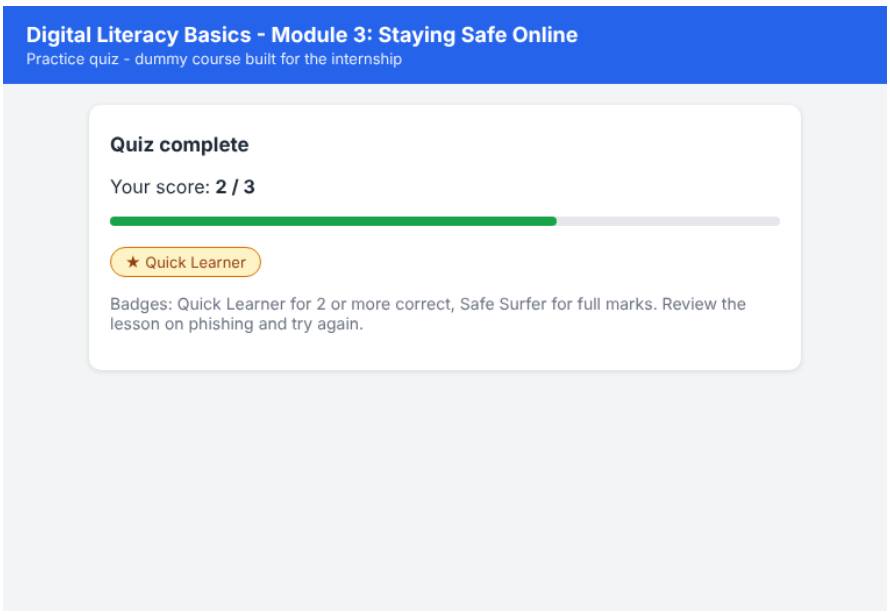


Figure 4: The result screen awards a badge for a score of 2 or more

The quiz is a single HTML page with a small JavaScript, built for learning, and was tested in a browser. It checks that an answer is selected before it is evaluated, shows the correct option in green and the wrong choice in red, explains the answer, and gives the badges by simple rules (Quick Learner for two or more correct, Safe Surfer for full marks). A bug found during testing - the badges were shown to every learner, even with one correct answer - was corrected so that they depend on the score.

3.6 Gamification in learning

Gamification is the use of game elements - points, badges, levels, leaderboards, challenges, progress bars and rewards - in a non-game setting to raise motivation. It works best when it supports real learning goals: points for completing practice, badges for mastery of a skill, and a progress bar that shows how far the learner has come. The main cautions are that a leaderboard may discourage the learners who are at the bottom (so it is better to compare with one's own earlier record), that rewards must not replace interest in the subject, and that the rules must be fair and simple. In the dummy pilot, 80 learners were divided into two groups of 40: one group received points and badges for completing lessons and quizzes, and the other did not. The completion rates are shown below.

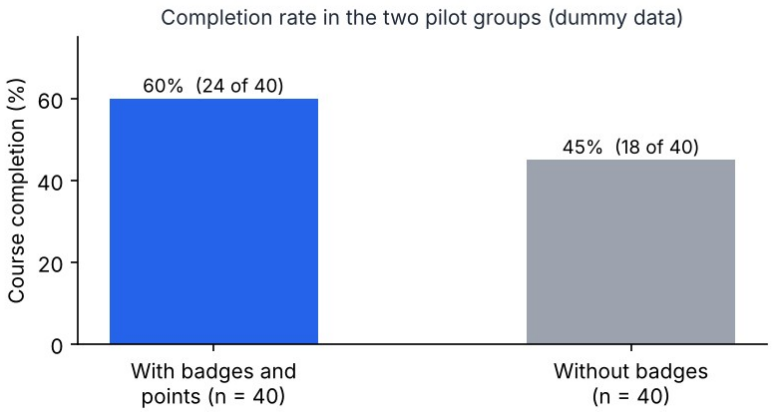


Figure 5: Completion in the gamified and the plain group (dummy pilot)

In the gamified group 24 of 40 learners (60%) completed the course, against 18 of 40 (45%) in the plain group, a difference of 15 percentage points. The student noted that with only 40 learners in each group the difference could arise by chance, that the groups were not randomly assigned in a real experiment, and that the data is dummy data, so the result only illustrates how such a comparison is made.

3.7 Student engagement methods

Engagement is the degree of attention, curiosity and effort that learners put into learning. Online, it falls quickly if the learner only watches and listens. Methods to raise it include starting with a question or a problem, polls and quick quizzes during the live class, short activities in every lesson, discussion questions in a forum, group tasks in breakout rooms, case studies from the learners' own life, personal feedback and encouragement, flexible deadlines for learners who have connectivity problems, and a clear schedule with reminders. The teacher's presence matters: a welcome message, a prompt reply to doubts and a weekly summary show learners that someone cares. Warning signs of low engagement include no login for several days, unfinished modules and no activity in the forum, and the teacher should contact such learners early with a friendly message and offer help.

3.8 Video lecture production

A video lecture should be short (six to ten minutes), focused on one outcome, and prepared with a script. The production steps are scripting (the opening, the main points, an example and a summary), storyboarding (a sketch of what will appear on screen), recording (a quiet room, a steady camera or screen recording, good light and clear audio - audio quality matters more than video quality), editing (cutting mistakes, adding titles, images and captions), publishing and checking the viewing and feedback. Good practice is to speak naturally and slowly, to use a visual for each idea, to show the main points on the screen, to add captions in the learners' language, and to place a question at the end. A script template and the production flow are shown below.



Figure 6: Steps of video lecture production

Time	On screen	Voice-over / action
0:00-0:30	Title and the question 'Is this SMS real?'	Greeting and the outcome of the lesson
0:30-2:30	Three warning signs, one by one	Explain each sign with an example
2:30-4:30	Two messages side by side	Ask the learner to pause and decide which one is fake, then reveal
4:30-5:30	Summary slide: Do not click, do not share OTP, call the bank	Repeat the three actions
5:30-6:00	Link to the quiz	Invite the learner to take the quiz

3.9 Learning analytics

Learning analytics is the collection and analysis of data about learners and their contexts, to understand and improve learning. An LMS records logins, time spent, pages viewed, quiz scores, assignment submission and forum use. This data is used to see which learners need help, which parts of the course are difficult, and whether a change has improved the results. Two analyses were made for the dummy course of 80 learners. The first is the progress of the learners through the modules, shown below.

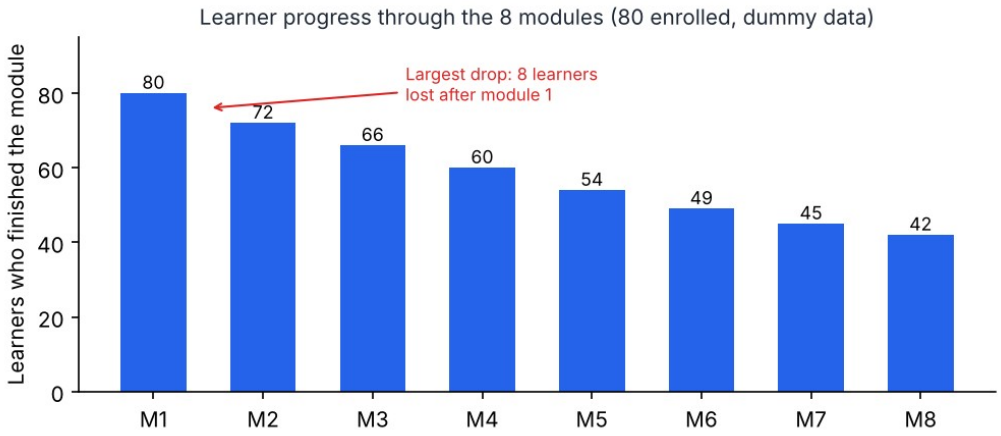


Figure 7: Learners who finished each module

The number of learners falls from 80 in the first module to 42 in the last, a completion rate of 42 / 80 = 52.5%. The largest single fall is after the first module, where 8 learners (10% of 80) stopped, followed by falls of 6 each after modules 2, 3 and 4. This suggests that the first module should be made easier and more welcoming, with a short orientation and a quick win, and that learners who do not log in for a week should get a reminder. The second analysis is the question-wise difficulty of the final quiz, taken by the 42 learners who finished. The difficulty index is the percentage of learners who answered a question correctly.

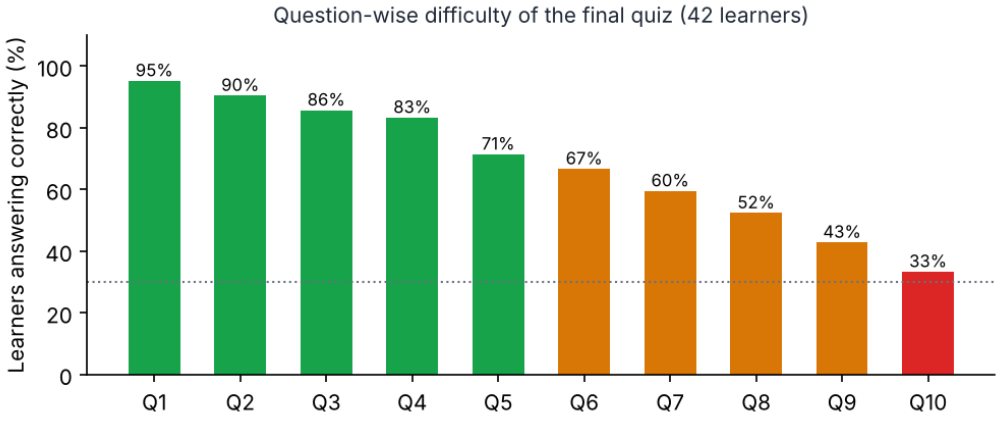


Figure 8: Difficulty of each question of the final quiz

The total of correct answers was 40 + 38 + 36 + 35 + 30 + 28 + 25 + 22 + 18 + 14 = 286, so the mean score is 286 / 42 = 6.81 out of 10, or 68.1%. Question 10, answered correctly by only 14 of the 42 learners (33%), was reviewed: it was found to be unclear because the stem had a double negative, and it was reworded. Question 1, answered correctly by 95%, is very easy and does not separate the learners, so it may be replaced by a more useful question. The student noted that data about learners is personal data and must be protected, used only for improving learning, and never to shame a learner.

3.10 Course design

Course design is the planning of a course so that the learners reach clear outcomes. A common model is ADDIE: analyse the learners and the need, design the outcomes, modules and assessment, develop the content and activities, implement the course on a platform, and evaluate and improve it using feedback and data. The principle of constructive alignment means that the outcomes, the learning activities and the assessments must match: if the outcome is 'identify a phishing message', the assessment must ask learners to identify such messages, and the activities must give practice in it. Outcomes are written with action verbs (identify, create, explain, apply) rather than 'understand'. A course map for the dummy course is shown below.



Figure 9: The ADDIE model of course design

Module	Outcome (the learner can ...)	Activity	Assessment
1 Computer basics	name the parts of a computer and use a mouse and keyboard	Video and screen exercise	5-question quiz
2 Internet and email	send an email with an attachment	Practice email task	Quiz and assignment
3 Staying safe online	identify a phishing message and make a strong password	Examples and discussion	Quiz with feedback (Figures 2-4)
4 Digital payments	make a UPI payment safely and check a transaction	Simulation with dummy app	Scenario quiz
5 Office tools	prepare a simple document and a table	Template exercise	Submitted file with rubric

The first five modules are shown; modules 6 to 8 covered online services, finding reliable information and a final project. The course was planned for four weeks, with about three hours of work in a week.

4. Week-wise Plan and Execution

Week	Days	Activities
Week 1	Days 1-6	Orientation and foundations: company induction, introduction to educational technology, the EdTech landscape and learning management systems.
Week 2	Days 7-12	Tools and content: online class tools, e-content creation and assessment tools.
Week 3	Days 13-18	Engagement: gamification in learning, student engagement methods and video lecture production.

Week	Days	Activities
Week 4	Days 19-24	Application: learning analytics, course design, the Digital Learning Content & LMS project, a case study exercise and a mock practical exercise.
Week 5	Days 25-28	Project closure: supervisor review and feedback, internship report drafting, presentation preparation and final review.

5. Week-wise Narrative

Week 1: The first week was devoted to orientation and foundations. After company induction, the student studied the types of EdTech tools and trends and learnt the main parts of an LMS by setting up a dummy course.

Week 2: In the second week the student practised live-class tools, wrote a micro-lesson plan and built a working quiz page with instant feedback.

Week 3: The third week was devoted to gamification, engagement methods and the scripting and storyboarding of a video lesson.

Week 4: In the fourth week the student analysed the dummy learner data, designed the course map, completed the project, a case study exercise and a mock practical exercise.

Week 5: The last week was devoted to project closure: supervisor review and feedback, drafting of the internship report, preparation of a presentation and final review before submission.

6. Topic-wise Learning in Detail

6.1 EdTech Landscape

Concept: Types of educational technology, platforms and current trends. **Practice carried out:** Making a chart of tools by purpose and writing the trends with examples.

6.2 Learning Management Systems

Concept: LMS features, course setup, enrolment and learner tracking. **Practice carried out:** Setting up the structure and completion rules of a dummy course (Figure 1).

6.3 Online Class Tools

Concept: Video conferencing, whiteboards and live-class tools. **Practice carried out:** Preparing a live-class plan with polls and a low-bandwidth option.

6.4 E-Content Creation

Concept: Preparing slides, notes, quizzes and digital learning material. **Practice carried out:** Writing a micro-lesson plan and designing five slides (section 3.4).

6.5 Assessment Tools

Concept: Online quizzes, tests, rubrics and result tracking. **Practice carried out:** Building a working quiz page and writing a rubric (Figures 2-4).

6.6 Gamification in Learning

Concept: Points, badges, challenges and game elements in learning. **Practice carried out:** Designing a badge system and comparing two groups (Figure 5).

6.7 Student Engagement Methods

Concept: Polls, discussions, activities and feedback to keep learners active. **Practice carried out:** Planning engagement activities for each module and a reminder message.

6.8 Video Lecture Production

Concept: Scripting, recording, editing and publishing video lessons. **Practice carried out:** Writing a six-minute script and storyboard (section 3.8).

6.9 Learning Analytics

Concept: Learner data, progress reports and performance trends. **Practice carried out:** Calculating completion and question difficulty and drawing charts (Figures 7 and 8).

6.10 Course Design

Concept: Learning outcomes, modules, content flow and assessment plan. **Practice carried out:** Preparing the course map with outcomes, activities and assessment (section 3.10).

7. Ethics, Privacy and Inclusion in EdTech - Points Studied

EdTech works with children and young people and with their data, so ethical use is essential. The points studied are summarised below.

- **Privacy of learners:** collect only the data needed, store it safely and do not share it; the Digital Personal Data Protection Act, 2023 applies, with special care for children.
- **Copyright:** use only original or properly licensed content and give credit.
- **Accessibility and inclusion:** use captions, readable fonts, low-bandwidth options and local languages so that all learners can take part.
- **Honest claims:** do not promise results, jobs or certificates that the course cannot give.

- **Healthy use:** plan breaks and limit unnecessary screen time.
- **Fair use of data:** use analytics to help learners, and not to rank or shame them.
- **Academic integrity:** teach learners to avoid copying and design assessments to test understanding.

8. Process Flows Studied

8.1 Steps of building an online course

- Define the learners, the need and the outcomes.
- Design the modules, activities and assessments, aligned with the outcomes.
- Prepare the content in short units with captions and practice.
- Set up the course in the LMS with completion rules, and test it on a phone and a laptop.
- Open it, watch the data and feedback, and improve.

8.2 Steps of a learning analytics review

- Export the learner data (logins, progress, scores) from the LMS.
- Remove personal identifiers wherever they are not needed.
- Calculate completion, time and the difficulty of questions.
- Find where the learners drop out and which items are unclear.
- Make changes, tell the learners and check the data again next time.

9. Worked Examples with Sample (Dummy) Data

Worked example 1 - completion and dropout. In the dummy course, 80 learners enrolled and 42 finished all the modules. The completion rate is 42 / 80 = 52.5% and the dropout is 100 - 52.5 = 47.5% (38 learners). After module 1 the number is 72, so 8 of the 80 (10%) left; the later falls are 6 (after module 2), 6 (after module 3), 6 (after module 4), 5 (after module 5), 4 (after module 6) and 3 (after module 7). If a welcome call and a first-week reminder could keep half of the 8 early leavers, the completion would rise by 4 learners to 46 / 80 = 57.5%.

Worked example 2 - effect size of gamification. The completion was 60% (24 of 40) with badges and 45% (18 of 40) without. The difference is 15 percentage points, and the relative increase is 15 / 45 = 33%. The standard error of the difference in two proportions is the square root of [0.60 x 0.40 / 40 + 0.45 x 0.55 / 40] = square root of (0.006 + 0.0061875) = about 0.110, so the 95% margin is about 1.96 x 0.110 = 0.215, or 21.6 points, which is larger than the 15-point difference. The student concluded that the sample is too small to say with confidence that badges helped, and that a larger test is needed.

Worked example 3 - estimating the time of video production. A rule of thumb used in the practice is that one minute of finished video takes about 20 to 30 minutes of work for scripting, recording and editing. A course with 24 lessons of six minutes (144 minutes of video) would therefore need 144 x 25 = 3,600 minutes, or 60 hours. This is why short videos, reuse of slides and a team approach are important.

10. Project: Digital Learning Content & LMS Project

Problem statement. A fictitious training organisation in Patna wants to offer an online course in basic digital literacy for first-generation learners, many of whom use only a smartphone. The organisation asks for a course design, a working sample assessment, a plan for the content and the video lessons, and a method to monitor learner progress. A dummy learner data set of 80 learners was used for the exercise.

10.1 Method

- Analyse the learners and write the outcomes for eight modules (section 3.10).
- Set up the course structure and completion rules in an LMS design (Figure 1).
- Write the micro-lessons and the video script and storyboard (sections 3.4 and 3.8).
- Build a working quiz page with instant feedback and badges (Figures 2-4).
- Analyse the dummy learner data and propose improvements (section 3.9).

10.2 Key findings

Area	Finding
Completion	80 enrolled, 42 completed (52.5%); the largest drop is after module 1 (10%).
Assessment	Mean score on the final quiz 6.81 out of 10 (68.1%); Question 10 was too hard (33% correct) and was reworded; Question 1 was too easy (95%).
Gamification	Completion 60% with badges against 45% without; the sample is too small for a firm conclusion.
Quiz page	Works on phone and laptop; one bug (badges given to everyone) found in testing and fixed.
Content	Eight modules with 3 to 4 micro-lessons of 5 to 7 minutes; videos with captions in Hindi and English.

10.3 Recommendations

- Add a short, friendly orientation and a first quick win in module 1 to reduce early dropout.
- Send reminders to the learners who have not logged in for seven days.

- Keep videos below eight minutes with captions and a low-bandwidth (audio and slides) version.
- Review the questions after every batch with the difficulty index and reword unclear ones.
- Run the badge test again with a larger group before deciding to use it throughout.

10.4 Limitations

The learner data is dummy data, the course was not run with real learners, no real LMS server was used (the structure was designed as it would be set up), and the pilot groups were small. A real project would require learner consent, data protection measures, accessibility testing and a pilot with real learners. The student noted that the learning lies in the method of design and the habit of using evidence.

11. Case Study Exercise: Moving a Tuition Centre Online During a School Closure

A fictitious tuition centre in Gaya with 120 students of classes 9 and 10 must move online for four weeks because of floods. Most students have only one smartphone in the family, shared with others, and the data plans are limited. The teacher asks the student for a plan.

Analysis. A survey of 60 families showed that 70% had a smartphone, but only 35% could attend a live video class for an hour at the fixed time, since the phone was shared or the network was weak. Many families could use WhatsApp and a few could use YouTube.

Plan. The student proposed a blended low-bandwidth plan: three short recorded videos of 8 minutes a day sent by a messaging group and also posted on a video platform at low resolution, with notes as small PDF files; a daily 20-minute live doubt session at two time slots with audio and slides; a weekly 10-question quiz built as a simple form with automatic marking; and a phone call from the teacher to the learners who had not responded for three days. Printed practice sheets were arranged for those with no phone.

Expected benefits and cautions. The plan reaches more learners than a one-hour live class and keeps the contact through calls. The cautions noted were data privacy (the phone numbers in a group must be protected and only the teacher should post), the need for parents' consent for recording the children, the limited ability to test understanding and the extra time for the teacher. The numbers in the case are illustrative.

Lessons. The best tool is the one that learners can actually use; technology must be chosen to fit the learners' conditions; and personal contact is part of good online teaching.

12. Mock Practical Exercise: Designing a Short Online Module

A mock practical exercise was conducted at the end of the learning phase to check that the student could design a complete short online module independently within a fixed time. The task was to design a module titled 'Making a good presentation' for college students. The steps followed were:

1. Learners and outcomes: the learners were described and three outcomes were written with action verbs.
2. Structure: three lessons of six minutes were planned with an activity and a quiz.
3. Content: slides and a one-page note were drafted for each lesson, with images from licensed sources.
4. Video: a script and storyboard were written for the first lesson.
5. Assessment: a five-question quiz with feedback and a rubric for a presentation task were written.
6. Engagement and gamification: a badge for each lesson and a forum question were added.
7. Analytics plan: the data to be checked (completion, scores and time) was listed.
8. Presentation: the module design was presented to the supervisor in about five minutes.

13. Sample Checklists Prepared

Course quality checklist

- Outcomes are clear and measurable; activities and assessments match them.

- Content is in short units with captions and readable design.
- Works on a phone; a low-bandwidth option exists.
- Sources are licensed and credited.

Learner data checklist

- Only needed data is collected; consent obtained.
- Data is stored safely and shared only with those who need it.
- Analytics is used to help learners.
- Learners know how their data is used.

14. Tools and Resources

The following tools and resources were used during the internship: a learning management system design (Moodle and Google Classroom features studied), a browser and a text editor for the quiz page, MS PowerPoint and Canva-type slide tools for e-content, screen recording and video editing tools, MS Excel and Python for learner analytics and charts, MS Word for the report, the DIKSHA and SWAYAM platforms for study, dummy course material and templates provided by the agency, and regular mentor feedback.

15. Programme at a Glance

The programme was in the domain of Educational Technology (EdTech) with RKNEXORA PRIVATE LIMITED as the agency. It required a minimum of 120 hours over 28 learning days, organised in three phases: Orientation (Days 1-2), Learning and Practice (Days 3-22) and Project and Review (Days 23-28). Attendance was recorded in the Internship Attendance Log, signed by the supervisor (see Attendance Sheet).

The approximate distribution of the effort across the topics is shown below.

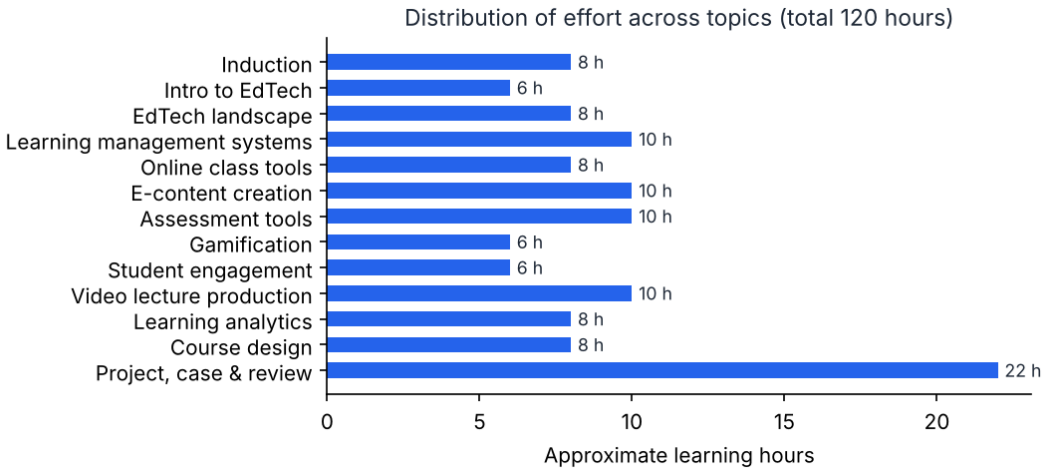


Figure 10: Distribution of learning effort across topics (120 hours)

LEARNING OUTCOME

a. Knowledge Acquired

1. The types of EdTech tools, platforms and current trends.
2. The features and setup of a learning management system.
3. Conducting effective live online classes.
4. Creating micro-lessons and accessible e-content.
5. Designing online quizzes, rubrics and feedback.
6. Gamification, its benefits and cautions.
7. Methods to keep learners engaged.
8. Scripting, recording and editing video lectures.
9. Reading learning analytics: completion and question difficulty.
10. Designing a course with ADDIE and constructive alignment.

Glossary of Terms Learned

- **EdTech** - Educational technology.
- **LMS** - Learning Management System.
- **Blended learning** - A mix of online and classroom learning.
- **Micro-learning** - Learning in very small units.
- **Formative / Summative assessment** - To help learning / to measure results.
- **Rubric** - A scoring guide with criteria and levels.
- **Gamification** - Using game elements in learning.
- **Engagement** - Attention and effort of learners.
- **Learning analytics** - Analysis of learner data to improve learning.
- **Completion rate** - Learners who finish divided by those enrolled.
- **Difficulty index** - Percentage answering a question correctly.
- **ADDIE** - Analyse, design, develop, implement, evaluate.
- **Constructive alignment** - Matching outcomes, activities and assessment.
- **OER** - Open educational resources.
- **Creative Commons** - A set of open licences.
- **Digital divide** - The gap between those who have and do not have access.
- **Synchronous / Asynchronous** - Live / at the learner's own time.
- **SWAYAM / DIKSHA** - National online learning platforms of India.
- **Adaptive learning** - Learning that adjusts to the learner.

b. Skills Learned

Technical skills. Setting up a course structure in an LMS; creating micro-lessons and slides; building quizzes and a quiz web page; scripting and storyboarding videos; designing gamification; calculating completion and difficulty indices; drawing charts of learner data; and designing a course.

Professional skills. Empathy for learners; clear and simple communication; creativity; planning and organisation; attention to detail; patience; collaboration; and the habit of testing with users and improving.

These skills were developed through repeated practice and daily documentation. Building the quiz page and testing it as a learner showed the value of user testing, analysing the dummy data built analytical habits, and presenting the course design to the supervisor built confidence.

c. Observed Attitude and Gained Values

1. **Learner first:** every design choice must be judged by how it helps the learner.
2. **Inclusion:** learners with weak connectivity, shared phones or different abilities must not be left out.
3. **Privacy and honesty:** only dummy data was used, and the importance of protecting learner data and of honest claims was understood.
4. **Continuous improvement:** use feedback and data to make the course better.
5. **Discipline and punctuality:** regular reporting at fixed timings and daily attendance records.

d. Challenges Faced

Understanding how an LMS organises courses was handled by designing the dummy course structure step by step.

Writing good quiz questions was handled by using real mistakes as wrong options and testing the questions with a few people.

Fixing the quiz page bug (badges for all) was handled by testing each path and linking the badges to the score.

Reading analytics without drawing wrong conclusions was handled by noting the sample size and the limits.

Designing for low bandwidth was handled by planning audio and slides options and small files.

Managing time between design, documentation and report writing was handled by dividing work week by week and using review days for corrections.

e. Career Pathways

The knowledge gained supports careers such as Instructional Design Assistant, E-Learning Content Developer, LMS Administrator, Online Course Coordinator, Academic Counsellor in EdTech, Learning Experience Executive, Video Lesson Producer and Learning Analytics Assistant, besides teaching and training roles that use technology. Opportunities exist in EdTech companies, schools, colleges and universities, coaching institutes, corporate training teams, NGOs working in education and government education projects. Related higher-study options include B.Ed., M.Ed., MA in education, BCA or MCA, MBA and certificate courses in instructional design, Moodle administration, video production and data analysis. A good next step is to build a small online course on a free LMS, publish it for a few learners and collect their feedback.

CONCLUSION

The internship in Educational Technology (EdTech) was a valuable learning experience that integrated academic knowledge with practical activity. The programme achieved its objectives through structured learning, guided practice, assignments, supervisor review and documentation.

The student gained a clear understanding of how technology is used to design, deliver and improve teaching and learning - through the study of the EdTech landscape, learning management systems, class tools, e-content, assessment, gamification, engagement, video production, learning analytics and course design - and acquired practical experience by designing a dummy online course with a working quiz page and a learner data analysis. The experience has strengthened the student's employability and provides a foundation for further study and work in education and technology.

Reflection

Looking back, the most useful lesson of the internship was that good EdTech begins with the learner and the outcome, and the tool comes after. The quiz page showed how a small piece of instant feedback can help a learner, and the bug in the badges showed that every path of a tool must be tested. The learner data showed where learners drop out, and the case of the tuition centre showed that the best technology is the one that the learner can actually use. The habits of testing, using evidence and respecting privacy apply to every educational project.

Objectives and Achievement

The objective of the EdTech landscape and LMS was achieved through the tool chart and the course structure. The objective of class tools and e-content was achieved through the live-class plan and the micro-lesson. The objective of assessment, gamification and engagement was achieved through the working quiz page, the badge design and the engagement plan. The objective of video production was achieved through the script and storyboard. The objective of analytics and course design was achieved through the completion and difficulty analysis and the course map.

Future Scope

The knowledge gained can support further study of education, instructional design and technology, and careers in e-learning and training. With the growth of smartphones, digital public platforms such as DIKSHA and SWAYAM and the focus of the National Education Policy 2020 on technology, there is a continuing need in Bihar and across India for people who can design learning that reaches every learner.

Suggestions for Improvement of the Programme

1. Provide access to a free LMS such as Moodle for hands-on course setup.
2. Include a short module on accessibility and low-bandwidth design.
3. Add a group project with a pilot involving real learners with consent.
4. Continue regular mentor feedback and progress reviews.
5. Encourage students to maintain detailed daily documentation.

References

1. Kolb, D. A. (1984). Experiential Learning: Experience as the Source of Learning and Development. Prentice Hall.
2. University Grants Commission (UGC). Internship guidelines for higher education institutions.
3. Ministry of Education, Government of India. National Education Policy 2020.

4. Mishra, S. and Sharma, R. C. Interactive Multimedia in Education and Training. Idea Group Publishing.
5. Clark, R. C. and Mayer, R. E. e-Learning and the Science of Instruction. Wiley.
6. Dick, W., Carey, L. and Carey, J. O. The Systematic Design of Instruction. Pearson.
7. Kapp, K. M. The Gamification of Learning and Instruction. Wiley.
8. Ministry of Education, Government of India. National Education Policy 2020.
9. Government of India. The Digital Personal Data Protection Act, 2023.
10. RK NEXORA internship programme handbook.

ATTENDANCE SHEET

The Internship Attendance Log issued by the IPO, with the supervisor's signature for each day, is to be attached on this page.

[Attach the signed Internship Attendance Log here]